



Solution strengthening of ductile refractory HfMo_xNbTaTiZr high-entropy alloys



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ABSTRACT

The crystal structure, microstructure, and mechanical properties of HfMo_xNbTaTiZr ($x \leq 1$) alloys were investigated by X-ray diffraction, scanning electron microscopy, and compression testing. All the as-solidified HfMo_xNbTaTiZr alloys possess a simple body-centered cubic structure. Lattice parameters of the HfMo_xNbTaTiZr alloys decrease with an increase in Mo concentration, but the corresponding densities show an opposite trend. The yield strength of the HfMo_xNbTaTiZr alloys increases from 1015 MPa to 1512 MPa as x increases from 0 to 1. The solid solution strengthening is linearly proportional to a Mo concentration suggesting that the strengthening effect from each Mo atom is additive. Except for the HfMoNbTaTiZr alloy, HfMo_xNbTaTiZr ($x < 1$) alloys exhibit excellent room-temperature plasticity ($> 50\%$). The material with $x=0.75$ is a new ductile refractory high-entropy alloy with promising mechanical properties.

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1. Introduction

There are several ways to strengthen materials including strain hardening, solid-solution strengthening, precipitation hardening, etc. [1]. Solid-solution strengthening plays an important role for high-entropy alloys (HEAs) [2–4] and especially for refractory HEAs with a body-centered cubic (BCC) structure. Refractory HEAs exhibit encouraging mechanical properties and can be potentially used in extremely high temperature applications [5–11]. For example, a BCC HfNbTaTiZr alloy possesses an excellent combination of high yield strength ($\sigma = 929$ MPa) and plasticity ($\epsilon > 50\%$) [7]. After an addition of a high-modulus refractory Mo element, the compression yield strength of an equiatomic HfMoNbTaTiZr alloy becomes equal to 556 MPa at 1200 °C, which is very high for superalloys [11]. However, its fracture strain value equal to 12% is not safe enough due to unavoidable impact loading of mechanical parts during dynamic operations. In this study, the effect of Mo addition on the strength and ductility of HfMo_xNbTaTiZr ($x \leq 1$) alloys is studied, and the appropriate amount of added Mo is determined. In addition, a proposed mechanism for the solid solution strengthening produced by Mo addition (from 0 at% Mo at $x=0$ to

16.7 at% Mo at $x=1$) is consistent with the relationship between a strengthening magnitude and a solute concentration [12–15].

2. Experimental procedures

HfMo_xNbTaTiZr HEAs were prepared by vacuum arc melting in a water-cooled copper crucible under argon atmosphere. All raw elemental materials had purities higher than 99.9 wt%. A melting procedure was repeated at least five times to improve the chemical homogeneity of alloys. Crystal structures were examined with an X-ray diffractometer (Shimadzu XRD-6000) operated at 30 kV and 20 mA with a scanning rate of 4°/min from 20° to 100°. The obtained cast ingots were cut and polished to be studied under a scanning electron microscope (SEM, JEOL JSM-5410). Chemical compositions of the resulted cast alloys were analyzed by SEM and energy dispersive spectrometry. Compression tests were conducted using cylindrical samples with diameters of 3 mm and heights of 6 mm by means of an Instron 4468 universal testing machine. The samples were tested at an initial strain rate of 10^{-3} /s.

3. Results and discussion

Fig. 1 shows the x-ray diffraction (XRD) patterns of the as-solidified

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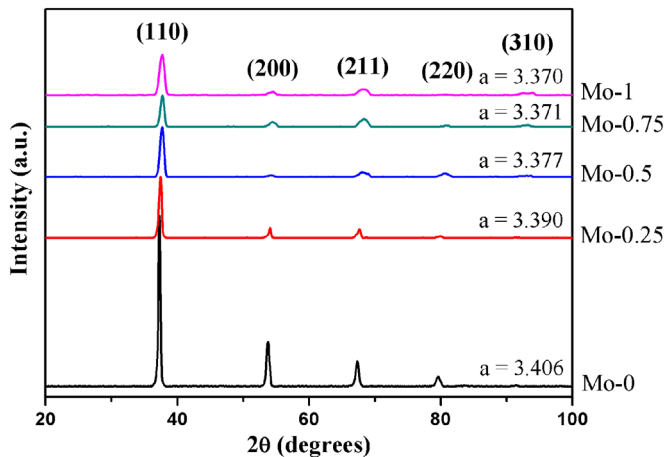


Fig. 1. X-ray diffraction patterns of the HfMo_xNbTaTiZr alloys.

HfMo_xNbTaTiZr alloys exhibiting a BCC structure. Table 1 lists the lattice constants, densities, and values of Vickers hardness of the same alloys. The lattice constant decreases with an increased Mo content due to a smaller value of the Mo lattice constant ($a_{\text{Mo}} = 3.147 \text{ \AA}$) than that of the base alloy ($a_{\text{HfNbTaTiZr}} = 3.406 \text{ \AA}$). However, the alloy density exhibits an opposite trend due to the higher density of Mo (10.22 g/cm^3) than that of the base alloy (9.91 g/cm^3). The Vickers hardness parameter increases with an increased Mo content, resulting in a total 50% increase of the Mo-1 alloy (at $x = 1$), which demonstrates that a large magnitude of the solid solution strengthening can be further enhanced for the base alloy by Mo addition (which will be later confirmed by the compression strength test results).

Fig. 2 shows the SEM backscattered images of the HfMo_xNbTaTiZr alloys. All alloys show the typical dendritic structure. Consistent with the XRD results, the obtained SEM data indicate no formation of other phases. The chemical compositions of dendrite and interdendrite regions in the HfMo_xNbTaTiZr alloys are listed in Table 2. The dendrite region is rich in the elements with higher melting points (Ta, Mo, and Nb). Increased proportion of Mo segregates to the dendrite in the alloy with higher Mo content, which further causes larger image contrast due to the average atomic number effect on brightness. In contrast, the interdendrite region is rich in the elements with lower melting points (Hf, Ti, and Zr). The interdendrite region is still characterized by a BCC structure, although the overall content of its three constituents with a hexagonal closed packed structure amounts up to 65 at% in the equiatomic HfMoNbTaTiZr alloy. This observation can be explained by a tendency of these three elements to form a BCC structure above the allotropic transition temperature, which means that the difference in free energy between the HCP and BCC structures is not large for these HCP elements as compared with that for Mo, Nb, and Ta.

Fig. 3(a) shows the compression stress-strain curves of the HfMo_xNbTaTiZr alloys. The obtained values of yield strength, fracture strain, and Young's modulus are listed in Table 3. It shows that the yield strength and Young's modulus values increase with an increased Mo content, and the obtained HfMo_xNbTaTiZr alloys (except for the Mo-1 alloy) have excellent compression plasticity, indicating no evidence of fracture at a compression strain of 50%.

The material with $x = 0.75$ is a new ductile refractory high-entropy alloy with promising mechanical properties.

Solid solution strengthening originates from several kinds of interactions between dislocations and solute atoms including elastic interaction (size effect), modulus interaction, chemical interaction, electrical interaction, and local order interaction. The elastic and modulus interactions dominate the solid solution strengthening [12]. For binary alloys, the yield strength increment ($\Delta\sigma$) of substitutional solid solution strengthening can be defined as

$$\Delta\sigma = \gamma Gc^n \quad (\text{MPa}) \quad (1)$$

where γ is the proportionality constant, G is the shear modulus, and c is the solute concentration. The value of $n = 1/2$ is observed for a number of conventional alloys. However, for some materials n is less than $1/2$ (but is normally higher than $1/3$), whereas for others n is greater than $1/2$ (but is normally less than unity) [12].

If the Mo atoms are considered as solute atoms, and the rest of the HfMo_xNbTaTiZr alloy as a solvent matrix, the HfMo_xNbTaTiZr alloys could be then regarded as quasi-binary alloys (HfNbTaTiZr)-Mo_x. The differences in lattice parameter ($a_{\text{Mo}} = 3.147 \text{ \AA}$; $a_{\text{HfNbTaTiZr}} = 3.406 \text{ \AA}$ from Table 1) and modulus ($E_{\text{Mo}} = 329 \text{ GPa}$; $E_{\text{HfNbTaTiZr}} = 85 \text{ GPa}$ from Table 3) between the Mo solute atoms and the HfNbTaTiZr matrix alloy cause the solid solution strengthening as suggested by a conventional solution strengthening mechanism. The modulus interactions are believed to be particularly important since the Young's modulus of Mo is more than three times greater than that of the HfNbTaTiZr alloy. The yield strength increase of the HfMo_xNbTaTiZr alloys as a function of Mo concentration is illustrated in Fig. 3 (b) indicating a linear dependence between these two parameters. The obtained correlation between the yield strength increase and Mo concentration can be expressed as

$$\Delta\sigma = 2921c \quad (\text{MPa}) \quad (2)$$

The resulting relationship does not match the $c^{1/2}$ dependence of the solid solution strengthening for conventional binary alloys. Fig. 3(b) also shows the yield strength increase of Al_xHfNbTaTiZr [16] and Mo_xNbTiVZr [9] HEAs as a function of Al and Mo concentrations, respectively. The Mo_xNbTiVZr alloys have a single solid solution phase at low Mo concentrations ($x \leq 20 \text{ at\%}$). However, the alloy system undergoes a phase transformation and becomes two phases at high Mo concentrations ($x > 20 \text{ at\%}$). In order to avoid the second phase strengthening effect, only the data points corresponding to low Mo concentrations were plotted. It should be noted that the data of the Al_xHfNbTaTiZr and Mo_xNbTiVZr alloys are also characterized by a linear correlation between the yield strength and solute concentration. This demonstrates that the strengthening of each solute atom in a high-entropy alloy matrix is additive.

4. Conclusions

HfMo_xNbTaTiZr alloys ($x \leq 1$) have a simple BCC structure. The lattice parameters of these alloys decrease with an increase in Mo concentration, but their densities show an opposite trend. The

Table 1
Lattice constant, density and Vickers hardness of the HfMo_xNbTaTiZr alloys.

Alloy	Mo-0	Mo-0.25	Mo-0.5	Mo-0.75	Mo-1
Lattice constant (Å)	3.406	3.390	3.377	3.371	3.370
Density (g/cm ³)	9.91	9.94	9.96	9.97	9.97
Vickers hardness (kg/mm ²)	335	395	480	492	505

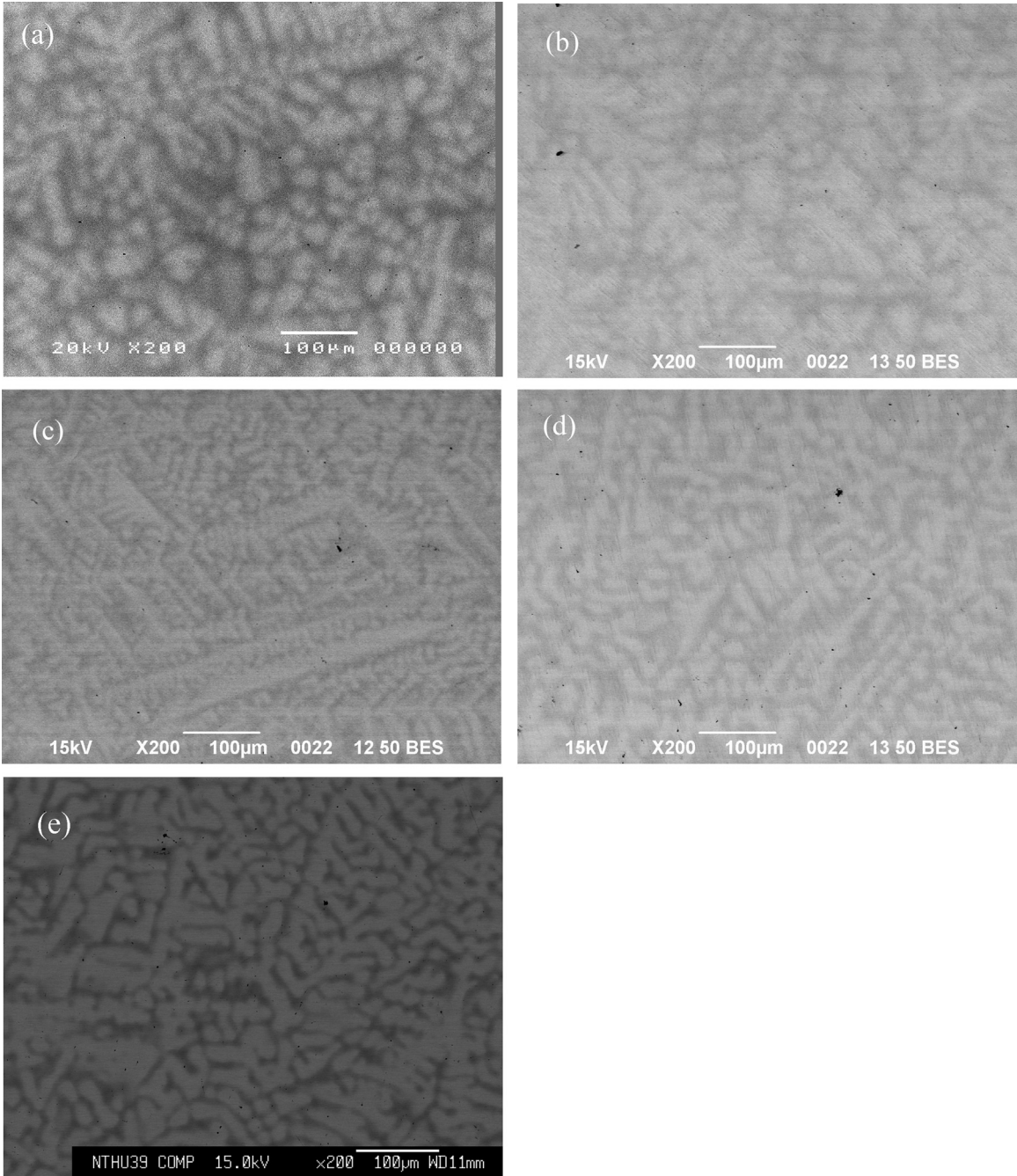


Fig. 2. Microstructures of the HfMo_xNbTaTiZr alloys: (a) x=0, (b) x=0.25, (c) x=0.5, (d) x=0.75, and (e) x=1.

Table 2
The chemical compositions of dendrite and interdendrite region in the HfMo_xNbTaTiZr alloys.

Alloy	Element	Hf	Mo	Nb	Ta	Ti	Zr
Mo-0	Nominal	20	–	20	20	20	20
	DR	19.9	–	21.7	24.4	16.6	17.5
	ID	22.2	–	16.5	15.9	21.7	23.6
Mo-0.25	Nominal	19.0	4.8	19.0	19.0	19.0	19.0
	DR	17.9	5.8	23.1	21.7	14.7	16.8
	ID	21.0	4.0	18.1	13.2	19.3	24.3
Mo-0.5	Nominal	18.2	9.1	18.2	18.2	18.2	18.2
	DR	15.5	9.3	20.7	26.3	15.4	12.9
	ID	21.7	8.7	14.5	11.1	19.3	24.8
Mo-0.75	Nominal	17.4	13.0	17.4	17.4	17.4	17.4
	DR	15.9	14.6	19.7	20.1	13.5	16.2
	ID	20.3	11.2	15.9	12.9	17.9	22.0
Mo-1	Nominal	16.7	16.7	16.7	16.7	16.7	16.7
	DR	14.3	18.4	19.5	24.4	12.4	10.8
	ID	21.1	13.6	12.3	9.9	18.3	24.7

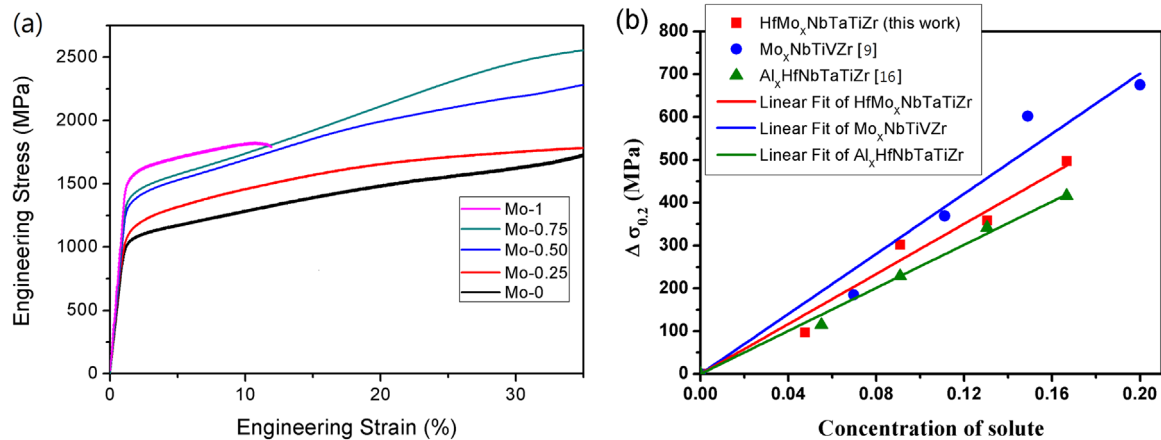


Fig. 3. The mechanical properties of HfMo_xNbTaTiZr alloys: (a) the engineering stress-strain curves, and (b) the increase of yield strength as functions of solute concentration in the HfMo_xNbTaTiZr and other high-entropy alloys.

Table 3

Compression yield strength, fracture strain, and Young's modulus of the HfMo_xNbTaTiZr alloys.

Code	Mo-0	Mo-0.25	Mo-0.5	Mo-0.75	Mo-1
Yield strength, $\sigma_{0.2}$ (MPa)	1015	1112	1317	1373	1512
Fracture strain, ϵ_f (%)	> 50	> 50	> 50	> 50	12
Young's modulus, E (GPa)	85	96	102	109	115

microstructures of all studied alloys correspond to a typical dendritic structure. No other phases were formed in the interdendrite region. The yield strength of the studied alloys increases from 1015 MPa to 1512 MPa with an increase in Mo concentration from zero to 16.7 at% indicating a significant solid solution strengthening in the HfMo_xNbTaTiZr alloys due to Mo addition. The obtained HfMo_xNbTaTiZr alloys show excellent room-temperature plasticity. The studied alloys, except for the Mo-1 alloy, underwent a 50% compression strain without any evidence of fracture. The HfMo_xNbTaTiZr alloys can be considered as quasi-binary alloys, and a linear correlation between their yield strength and a solute atom concentration is established.

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